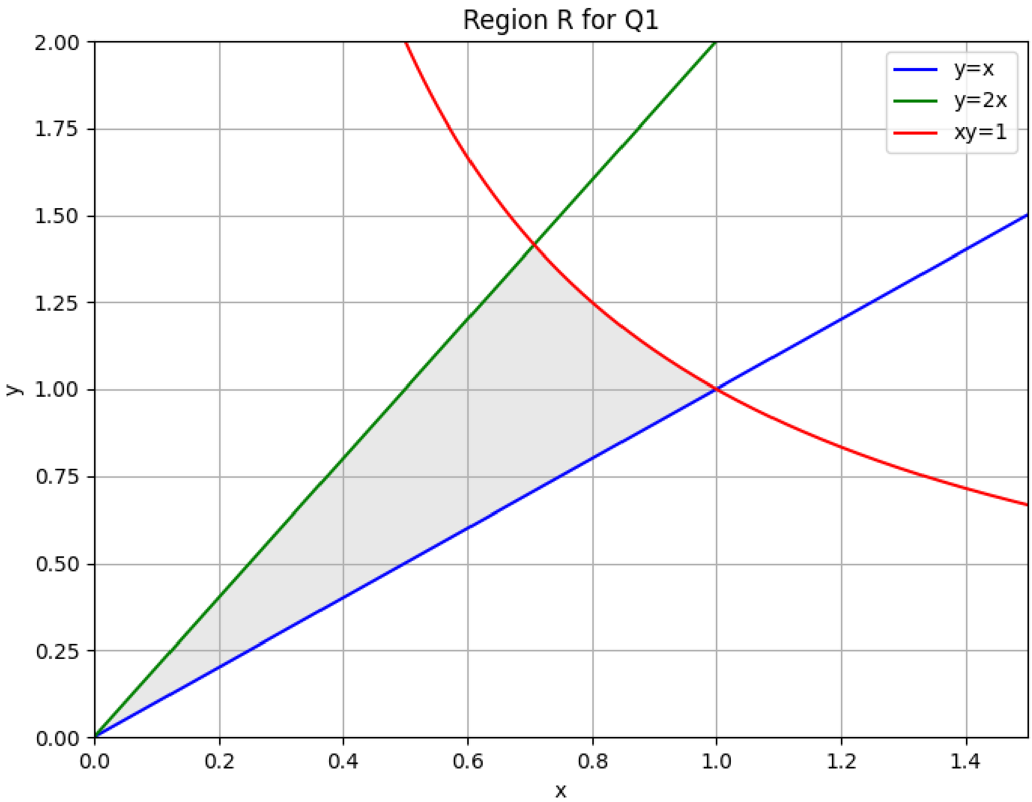
Q1.



Region R for Q1
y=x
y=2x
xy=1
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
y
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
x

Q1 $\iint_R x^2\,dA$

$y = 2x \qquad xy = 1$

$x(2x) = 1$

$2x^2 = 1$

$\sqrt{x^2} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$

$y = x \qquad xy = 1$

$x(x) = 1 \Rightarrow x^2 = 1$

$x = 1$

$$\int_0^{\frac{1}{\sqrt{2}}}\int_x^{2x} x^2\,dy\,dx + \int_{\frac{1}{\sqrt{2}}}^{1}\int_x^{1/x} x^2\,dy\,dx$$

$$\int_0^{\frac{1}{\sqrt{2}}} \left[x^2 y\right]_x^{2x} dx + \int_{\frac{1}{\sqrt{2}}}^{1} \left[x^2 y\right]_x^{1/x} dx$$

$$\int_0^{\frac{1}{\sqrt{2}}} x^2(2x-x)\,dx + \int_{\frac{1}{\sqrt{2}}}^{1} x^2\left(\frac{1}{x}-x\right)dx$$

$$\int_0^{\frac{1}{\sqrt{2}}} x^3\,dx + \int_{\frac{1}{\sqrt{2}}}^{1} x-x^3\,dx.$$

$$\left[\frac{x^4}{4}\right]_0^{\frac{1}{\sqrt{2}}} + \left[\frac{x^2}{2}-\frac{x^3}{3}\right]_{\frac{1}{\sqrt{2}}}^{1}$$

$$= \cancel{\frac{1}{16}}$$

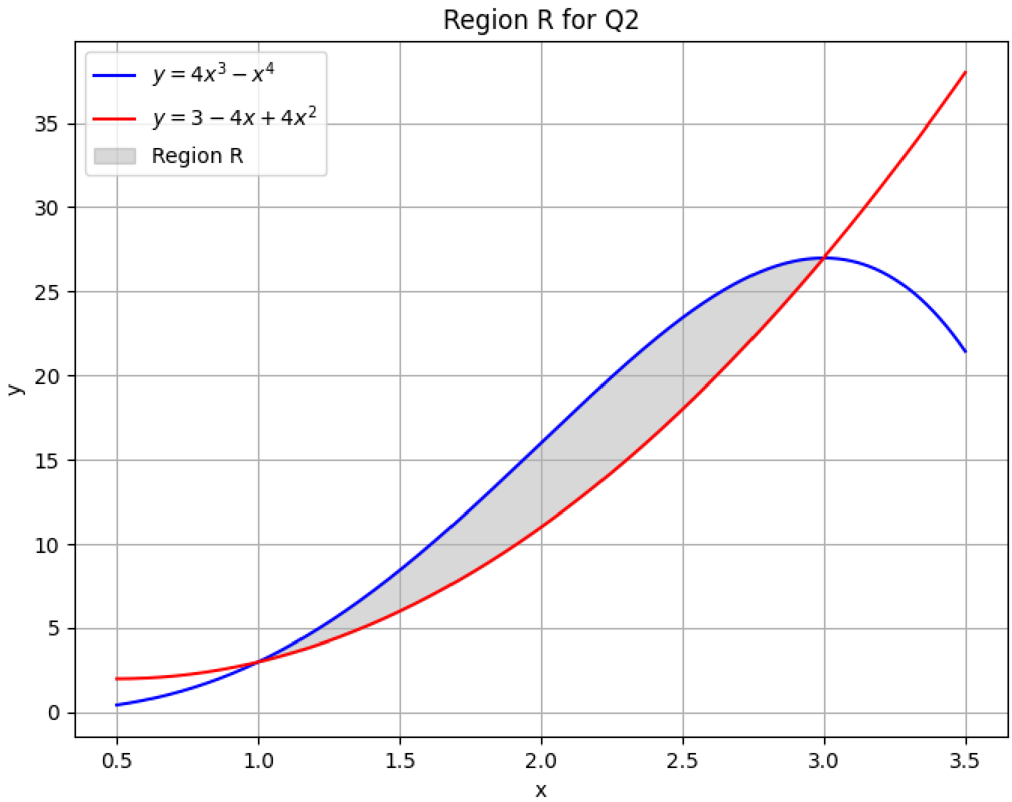
$$= \frac{1}{4}\left(\frac{1}{\sqrt{2}}\right)^4 - 0 + \frac{1}{2} - \frac{1}{4} + \frac{(1/\sqrt{2})^2}{2} - \frac{(1/\sqrt{2})^4}{4}$$

$$= \frac{1}{16} + \frac{1}{16} = \frac{2}{16}$$

$$= \frac{1}{8}$$

Q2.

(a)


Region R for Q2
$y = 4x^3 - x^4$
$y = 3 - 4x + 4x^2$
Region R
35
30
25
20
15
10
5
0
y
0.5
1.0
1.5
2.0
2.5
3.0
3.5
x

(b)

Q2b.

$$y = 4x^3 - x^4$$

$$y = 3 - 4x + 4x^2.$$

$$3 - 4x + 4x^2 = 4x^3 - x^4$$

$$\cancel{4x^3 - x^4 - 4x^2 + 4x + 3 = 0}$$

$$x^4 - 4x^3 + 4x^2 - 4x + 3 = 0$$

try $x = 1$. (root).

$$(1)^4 - 4(1)^3 + 4(1)^2 - 4(1) + 3 = 0$$

$$0 = 0$$

$x = 3$ (root)

$$(3)^4 - 4(3)^3 + 4(3)^2 - 4(3) + 3 = 0$$

$$0 = 0$$

when $x = 1$, $y = 4(1)^3 - (1)^4 = 3$

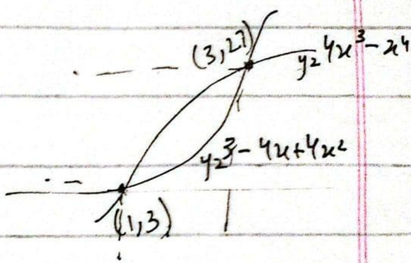
when $x = 3$, $y = 3 - 4(3) + 4(3)^2 = 27$

Intersection point. $(1, 3)$ $(3, 27)$

(c)

Intersection point. $(1,3)$ $(3,27)$

Q2c Find $\iint_R x \, dA$.



$$\int_1^3 \int_{3-4x+4x^2}^{4x^3-x^4} x \; dy \, dx$$

$$\int_1^3 xy \Big|_{3-4x+4x^2}^{4x^3-x^4} dx = \int_1^3 x(4x^3-x^4) - x(3-4x+4x^2) \, dx$$

$$= \int_1^3 4x^4 - x^5 - 3x + 4x^2 - 4x^3 \, dx$$

$$= \left[\frac{4x^5}{5} - \frac{x^6}{6} - \frac{3x^2}{2} + \frac{4x^3}{3} - \frac{4x^4}{4} \right]_1^3$$

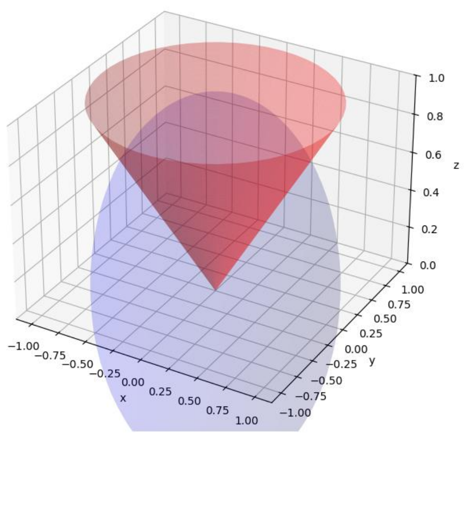
$$= \left(\frac{4(3)^5}{5} - \frac{(3)^6}{6} - \frac{3(3)^2}{2} + \frac{4(3)^3}{3} - (3)^4 \right)$$

$$- \left(\frac{4}{5} - \frac{1}{6} - \frac{3}{2} + \frac{4}{3} - 1 \right)$$

$$= \frac{224}{15} = 14.9 \approx 15$$

Q3.

Solid bounded by Sphere and Cone



Q3

$x = r\cos\theta$

$y = r\sin\theta$ $\qquad x^2+y^2 = r^2$

$z = z$

$x^2+y^2+z^2 = 1$

$r^2 + z^2 = 1$

Sphere $\quad z = \sqrt{1-r^2}$

Cone $\quad z = \sqrt{x^2+y^2} \quad z = \sqrt{r^2} \quad z = r$

$r = \sqrt{1-r^2}$

$r^2 = 1-r^2$

$2r^2 = 1$

$r^2 = \frac{1}{2}$

$r = \frac{1}{\sqrt{2}}$

$$V = \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\int_r^{\sqrt{1-r^2}} r \, dz\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} r[z]_r^{\sqrt{1-r^2}} \, dr\, d\theta = \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} r(\sqrt{1-r^2}-r)\, dr\, d\theta$$

$u = 1-r^2$

$du = -2r\,dr \Rightarrow -\frac{1}{2}du = r\,dr$

$$\int_0^{2\pi} -\frac{1}{2}\int_0^{\frac{1}{\sqrt{2}}} u^{1/2}\, du = -\frac{1}{2} \times \frac{2}{3} u^{3/2} = -\frac{1}{3}(1-r^2)^{3/2}$$

$$\int_0^{2\pi}\left(-\frac{1}{3}(1-r^2)^{3/2} - \frac{r^3}{3}\right]_0^{1/\sqrt{2}}$$

$$\int_0^{2\pi} \frac{-1}{3}\left(1-\frac{1}{2}\right)^{3/2} - \frac{1}{3}\left(\frac{1}{\sqrt{2}}\right)^3 - \left(-\frac{1}{3}(1)^{3/2} - 0\right)$$

$$\int_0^{2\pi} -\frac{1}{3}\left(\frac{1}{2\sqrt{2}}\right) - \frac{1}{3}\left(\frac{1}{2\sqrt{2}}\right) - \left(-\frac{1}{3}\right).$$

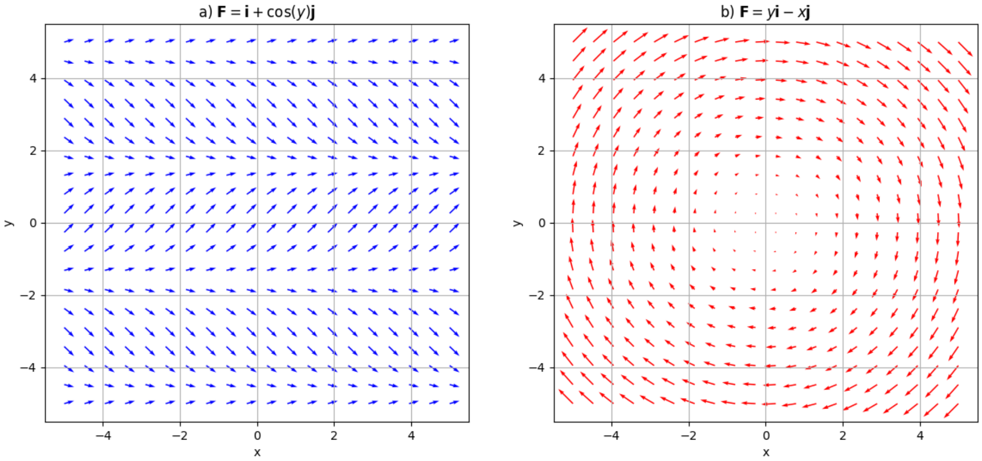
$$\int_0^{2\pi} \frac{-2}{6\sqrt{2}} + \frac{1}{3}$$

$$\int_0^{2\pi} -\frac{1}{3\sqrt{2}} + \frac{1}{3}\, d\theta = \left[\left(\frac{-1}{3\sqrt{2}} + \frac{1}{3}\right)\theta\right]_0^{2\pi}$$

$$= 2\pi\left(\frac{1}{3} - \frac{1}{3\sqrt{2}}\right) - 0$$

$$V = 0.61.$$

Q4.


a) $\mathbf{F} = \mathbf{i} + \cos(y)\mathbf{j}$
b) $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$
y
x
4
2
0
−2
−4

Q5.

Q5 $\int_C x^2 z\,dx - yx^2\,dy + 3\,dz$

C_1 $(0,0,0) \to (1,1,0)$

$x = t$ $\quad y = t$ $\quad z = 0$

$dx = dt$ $\quad dy = dt$ $\quad dz = 0$

$$\int_{C_1} x^2 z\,dx - yx^2\,dy + 3\,dz$$

$$\int_0^1 t^2(0)dt - t(t)^2\,dt + 3(0)$$

$$\int_0^1 0 - t^3\,dt + 0 = \left[\frac{-t^4}{4}\right]_0^1 = \frac{-1}{4} - 0$$

$$= \frac{-1}{4}$$

C_2 $(1,1,0) \to (1,1,1)$

$x = 1$ $y = 1$ $z = t$

$dx = 0$ $dy = 0$ $dz = dt$

$$\int_0^1 (1)^2 t(0) - (1)(1)^2(0) + 3(dt)$$

$$\int_0^1 0 - 0 + 3dt = \left[3t\right]_0^1 = 3 - 0 = 3$$

C_3 $(1,1,1) \to (0,0,0)$

$x = t$ $y = t$ $z = t$

$dx = dt$ $dy = dt$ $dz = dt$

$$\int_1^0 t^2(t)(dt) - t(t)^2 dt + 3(dt)$$

$$\int_1^0 \cancel{t^3\,dt} - \cancel{t^3\,dt} + 3\,dt$$

$$\int_1^0 3dt = \left[3t\right]_1^0 = 0 - 3 = -3$$

$$I = C_1 + C_2 + C_3$$

$$= -\frac{1}{4} + \cancel{3} - \cancel{3}$$

$$= -\frac{1}{4}$$

Q6.

Q6 $F = xy^2 i + x^2 y j$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

$2xy = 2xy$ hence $\vec{F}$ is conservative.

$\int F \cdot dr.$ $\quad \vec{F} = \nabla \phi = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \right)$

$\vec{F} = \cancel{f = xy^2}\ f(x,y)\hat{i} + g(x,y)\hat{j}$

$f(x,y) = \frac{\partial \phi}{\partial x}$ $\qquad g(x,y) = \frac{\partial \phi}{\partial y}$

$\int \frac{\partial \phi}{\partial x} = \int xy^2$ $\qquad \frac{\partial \phi}{\partial y} = x^2 y$ — (2)

$\phi = \int xy^2 \, \partial x$

$\phi = \frac{y^2 x^2}{2} + k(y)$

$\frac{\partial \phi}{\partial y} = \frac{x^2}{\cancel{2}} \times \cancel{2} y + k'(y)$

$\frac{\partial \phi}{\partial y} = x^2 y + k'(y)$ — (1)

Equal (1) = (2)

$x^2 y = x^2 y + k'(y)$

$0 = k'(y)$ $\qquad k(y) = \int k'(y) = \cancel{k(y)} \int 0 \, dy = 0$

$\phi = \frac{x^2y^2}{2} + 0$ $\quad$ $\phi(1,1)$ $\quad$ $\phi(0,0)$

$w = \int f \cdot dr = \phi(0,0) - \phi(1,1)$

$= \frac{(0)^2(0)^2}{2} - \frac{(1)^2(1)^2}{2}$

$w = -\frac{1}{2}$